WEEK 1 ASSIGNMENT

Cyber Security Fundamentals & Network Basics

Case Study Platform:**College / UniversityWebsite**

Marks: 100 | Duration: 1 Week (7 Days)

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Submission Date	07/11/2026

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Case Study Overview: The College / University Website

For this assignment I have chosen a **College / Universitywebsite** as the platform for all examples. A modern college website is much more than a brochure. It is a collection of connected online services that hold the personal, academic and financial data of thousands of students, teachers and staff. This makes it an attractive target for attackers and an excellent example for learning cyber security and networking.

Typical components of a college web platform

Component	What it does	Sensitive data involved
Public website	News, courses, notices, contact details	Mostly public, but defacement harms reputation
Admission portal	Online applications and counselling	Identity documents, marks, phone numbers
Student ERP / LMS	Attendance, assignments, e-learning	Login credentials, grades, personal profile
Fee payment gateway	Online payment of fees	Card / UPI / bank details
Results portal	Exam results and marksheets	Roll numbers, marks
Campus Wi-Fi and LAN	Internet for labs, library, hostels	All user traffic on the network
College email and cloud storage	Communication and files	Official communication, research data

Why this case study fits the assignment

- **Task 1 and 2:** the CIA triad and real threats (phishing, malware, DoS) can be shown with college examples.
- **Task 3 and 4:** the campus network (LAN), ISP link (WAN) and the journey of a request to the website are easy to explain.
- **Task 5:** firewalls, VPN, encryption and authentication all protect the portals above.
- **Task 6:** the IP address practical is carried out on my own system connected to a home / campus network.

Task 1: Introduction to Cyber Security (20 Marks)

1.1 Definition of Cyber Security

Cyber security is the practice of protecting computers, networks, servers, mobile devices, applications and data from unauthorised access, attacks, damage or theft. In my own words, it is like locking the doors and windows of our digital world: only the right people should be able to enter, the information inside should not be changed by outsiders, and the services should keep working whenever genuine users need them.

Example: when a student logs in to the college portal with a password and an OTP, the HTTPS lock in the browser encrypts the data, and the server's firewall blocks suspicious traffic, all of these are cyber security measures working together.

1.2 Importance of Cyber Security

- **Protects sensitive data:** student records, marks, Aadhaar / ID details, bank and card information must stay private.
- **Prevents financial loss:** cyber attacks such as ransomware and online fraud cost individuals and organisations large amounts of money every year.
- **Builds trust and protects reputation:** a hacked college website or leaked results quickly damage the trust of students and parents.
- **Ensures continuity of services:** admissions, online classes and exams depend on systems being available without interruption.
- **Supports legal compliance:** laws such as India's Information Technology Act, 2000 and the Digital Personal Data Protection Act, 2023 require organisations to safeguard personal data.
- **Protects national security and critical infrastructure:** power grids, hospitals, transport and government services depend on secure networks.
- **Safeguards personal privacy and safety:** prevents identity theft, stalking, blackmail and online harassment.

1.3 Goals of Cyber Security: The CIA Triad

The **CIA Triad** is the foundation model of information security. Every security control is designed to protect at least one of its three goals.

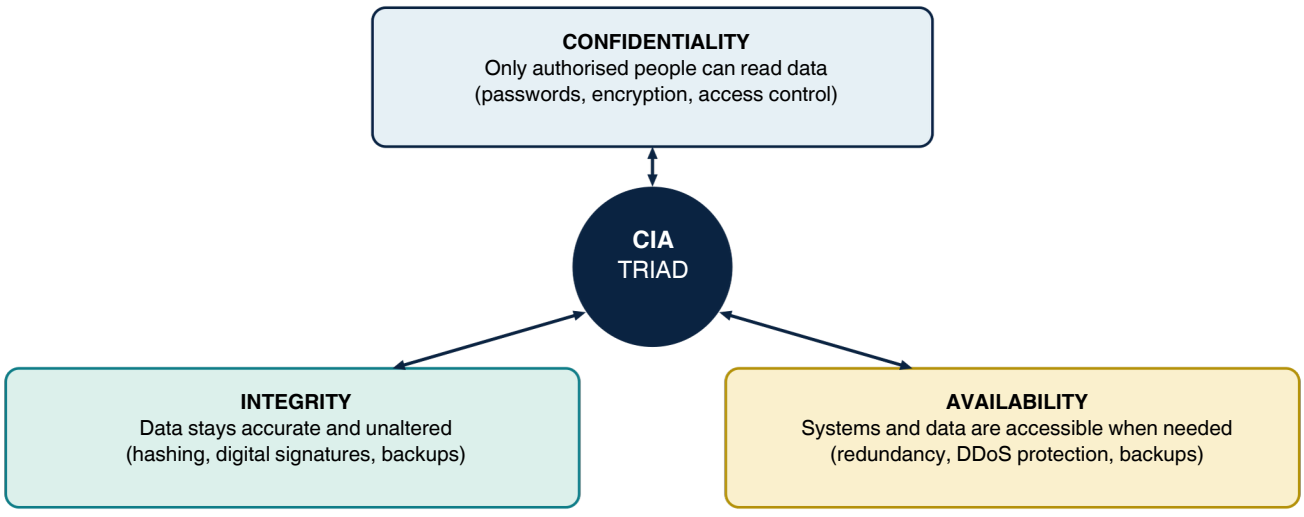


Figure 1: The CIA Triad - Confidentiality, Integrity, Availability

Goal	Meaning	College website example	Common controls
Confidentiality	Information is accessible only to authorised users.	Only the student and the examiner can view a particular student's marks.	Passwords, MFA, encryption, role-based access
Integrity	Data remains accurate, complete and unmodified.	No one can secretly change grades or attendance in the database.	Hashing, digital signatures, audit logs, backups
Availability	Systems and data are usable when required.	The results portal stays online on result day and the LMS is reachable during exams.	Load balancing, redundancy, DDoS protection, backups

1.4 Five Real-Life Applications of Cyber Security

No.	Area	How cyber security is applied
1	Banking and online payments	Encryption (HTTPS/TLS), OTP and two-factor authentication, fraud detection systems and secure UPI / card transactions protect money and account details.
2	Healthcare	Electronic health records are protected with access control and encryption so that patient data is not leaked or altered, and medical devices are kept secure.
3	E-commerce and online shopping	Secure payment gateways, PCI-DSS compliance, bot and fraud detection and account protection keep customers' data and orders safe.
4	Government and defence	Protection of citizen databases, e-governance services, DigiLocker and critical infrastructure against espionage and cyber warfare; agencies such as CERT-In respond to incidents.
5	Education (colleges and universities)	Securing student portals, online exams, research data and campus Wi-Fi with firewalls, authentication, antivirus and regular backups.

Task 2: Types of Cyber Threats (15 Marks)

Acyber threat is anypossible malicious act thattries to damage data, steal information or disrupt digital systems. I have researched and explained five common threats: **Malware, Virus, Phishing, Spyware and DoS Attack**. Each is explained with its definition, working and prevention methods, along with a college-related example.

2.1 Malware

Definition	Malware (short for **malicious software**) is an umbrella term for any program designed to harm, exploit or gain unauthorised access to a computer, network or data. Viruses, worms, trojans, ransomware and spyware are all types of malware.
How it works	• Reaches the victim through infected email attachments, pirated / cracked software, malicious websites, fake apps and infected USB drives. • Once the user runs the file, the malware installs itself, often hiding from the user. • It then performs its payload: stealing data, deleting or encrypting files, giving the attacker remote control, showing ads or installing more malware. • Infected machines can be joined to a botnet and used to attack others.
Prevention methods	• Install reputable antivirus / anti-malware and keep it updated. • Keep the operating system and applications patched. • Download software only from official sources; avoid cracked software. • Do not open unknown attachments or click unknown links. • Use standard (non-admin) user accounts for daily work. • Take regular offline / cloud backups.
College example	A student installs a cracked version of design software on a lab computer; hidden malware then spreads across the lab network.

2.2 Virus

Definition	A computer virus is a self-replicating piece of malicious code that **attaches itself to a legitimate file or program** and spreads when that infected file is opened or shared. Like a biological virus, it needs a host to spread.
How it works	• The virus attaches to a host such as an .exe file, a document with macros or the boot sector of a drive. • When the infected host is executed, the virus code runs and copies itself into other files or drives. • It spreads by USB pen drives, email attachments, shared folders and downloads. • Its payload may corrupt or delete files, slow down the system or crash programs.
Prevention methods	• Enable real-time antivirus scanning and scan USB drives before opening them. • Disable macros in documents from unknown sources. • Avoid unknown files, email attachments and pirated media. • Apply security patches regularly. • Keep regular backups so files can be restored.
College example	A pen drive passed between classmates carries a virus that infects assignment files on every computer it is plugged into.

2.3 Phishing

Definition	Phishing is a social-engineering attack in which the attacker sends fraudulent emails, SMS, calls or fake websites that **pretend to come from a trusted source** in order to trick the victim into revealing passwords, OTPs, card details or installing malware.
How it works	• The attacker sends an urgent message, for example "Your exam fee payment failed. Click here to pay or your hall ticket will be blocked." • The link opens a look-alike copy of the college login or payment page hosted on a fake domain. • The victim enters credentials, which are sent to the attacker. • The attacker then logs in as the victim or sells the data. Variants: spear phishing (targeted), smishing (SMS), vishing (voice call), whaling (executives).
Prevention methods	• Check the sender's address and hover over links to verify the real URL before clicking. • Never share OTP, PIN or passwords with anyone. • Enable multi-factor authentication (MFA) on all important accounts. • Use spam / email filters and keep the browser updated. • Verify unexpected requests through the official website or phone number. • Report cyber fraud at cybercrime.gov.in or helpline 1930 (India).
College example	Students receive a fake "scholarship verification" email that links to a copy of the college portal login page.

2.4 Spyware

Definition	Spyware is malicious software that **secretly monitors a user's activity** and sends the collected information, such as keystrokes, browsing history, screenshots and login details, to an attacker without the user's knowledge.
How it works	• It is bundled with free software, hidden inside fake apps / browser extensions or delivered through phishing attachments. • It runs silently in the background and may include a keylogger, screen capturer or microphone / camera access. • The stolen data is periodically sent to a remote server controlled by the attacker. • The attacker uses it for identity theft, bank fraud or blackmail.
Prevention methods	• Install apps and extensions only from official stores and check the permissions they ask for. • Use antivirus / anti-spyware tools and run periodic scans. • Keep the OS, browser and apps up to date. • Use a firewall to block suspicious outgoing connections. • Watch for warning signs: slow device, high data usage, unknown apps, battery drain.
College example	A free "exam notes" app requests access to the keyboard and contacts, and silently sends the student's ERP password to an attacker.

2.5 DoS Attack (Denial of Service)

Definition	A Denial of Service (DoS) attack tries to make a website, server or network **unavailable to genuine users** by overwhelming it with a flood of traffic or requests. When many compromised computers (a botnet) attack together, it is called a **DDoS** (Distributed Denial of Service) attack.
How it works	• The attacker sends a huge number of requests such as SYN floods, HTTP floods or UDP / ICMP floods to the target. • The server's bandwidth, CPU or memory is exhausted trying to answer fake requests. • Genuine users see slow loading or error pages, and the service may crash completely. • It attacks the **Availability** part of the CIA triad.
Prevention methods	• Use firewalls, intrusion prevention systems and rate limiting to drop excessive requests. • Use a CDN / DDoS protection service and load balancers. • Keep spare bandwidth and redundant servers. • Monitor traffic to detect unusual spikes early. • Prepare an incident response plan and contact the ISP quickly.
College example	An attacker floods the admission counselling portal so that real applicants cannot register before the deadline.

2.6 Summary Comparison

Threat	Main target	Attacker's goal	Key defence
Malware	Any device / network	Damage, control, steal data	Antivirus, patching, safe downloads
Virus	Files and programs	Spread and corrupt files	Real-time scanning, scan USB drives
Phishing	Human users	Steal credentials / money	Awareness, MFA, verify links
Spyware	User privacy	Secret surveillance	Official apps, anti-spyware
DoS / DDoS	Servers and networks	Make service unavailable	Firewall, CDN, rate limiting

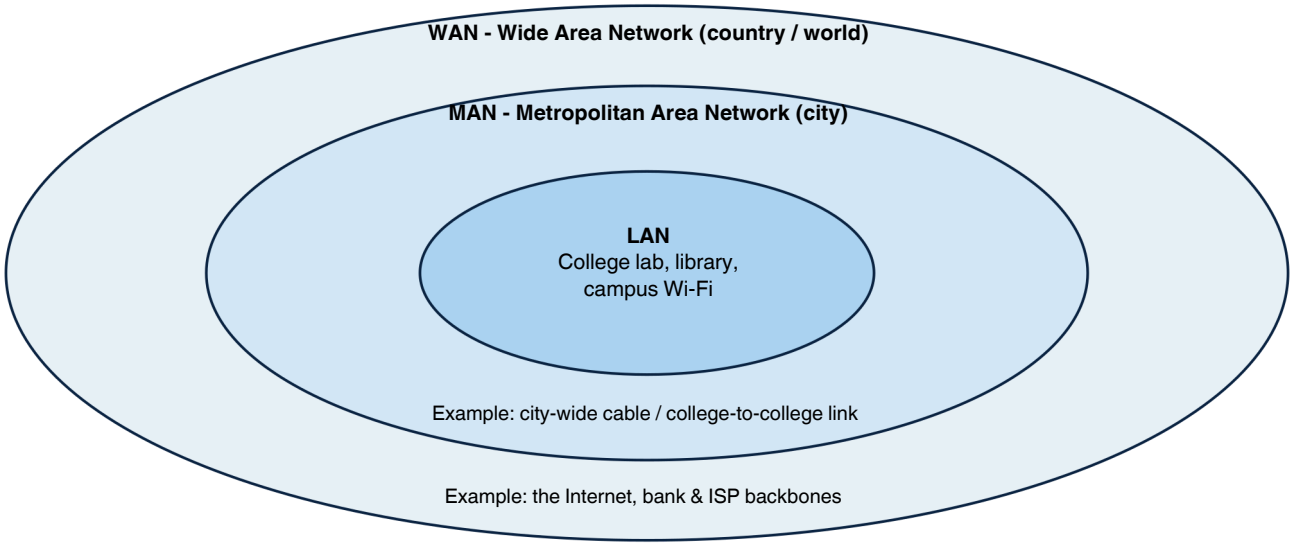
Task 3: Network Basics (20 Marks)

3.1 What is a Computer Network?

A computer network is a group of two or more computers and other devices (printers, servers, phones, etc.) that are **connected together by wired or wireless links** so that they can communicate, share data and share resources such as files, printers and an Internet connection. Every network has four basic elements: **nodes** (devices), **links** (cables / wireless), **networking devices** (router, switch, modem) and **protocols** (rules such as TCP/IP, HTTP, DNS).

Example: in a college, the computers in the lab, the library, the staff room and the Wi-Fi access points are connected together and to the Internet so that students can use the ERP, e-library and online classes.

3.2 Types of Networks: LAN, MAN, WAN



Feature	LAN	MAN	WAN
Full form	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	A room, building or campus (up to a few km)	A city or large town (about 5-50 km)	Countries / continents; worldwide
Speed	Very high	High	Lower than LAN, varies widely
Ownership	Usually a single organisation	A city authority or an ISP	Multiple organisations / ISPs
Cost	Low	Medium	High
Example	College computer lab or campus Wi-Fi	City-wide cable TV / broadband or linking two campuses of one university	The Internet

3.3 IP Address

An **IP (Internet Protocol) address** is a unique numerical label given to every device on a network. It identifies the device and tells other devices where to send data, just like a postal address tells the postman where to deliver a letter.

- **IPv4:** 32-bit address written as four numbers (0-255) separated by dots, e.g. 192.168.1.10.
- **IPv6:** 128-bit address written in hexadecimal, e.g. 2001:db8::1; created because IPv4 addresses are running out.

- An IP address can be **private** (inside a LAN) or **public** (on the Internet), and **static** or **dynamic** (explained in Task 6).

3.4 MAC Address

A **MAC (Media Access Control) address** is a 48-bit hardware address permanently assigned to a device's network interface card (NIC) by the manufacturer. It is written as six pairs of hexadecimal digits, for example 00:1A:2B:3C:4D:5E. The first half identifies the manufacturer (OUI) and the second half is a unique serial number. MAC addresses work at the Data Link layer and are used to deliver data to the correct device **inside the same local network**.

Point	IP Address	MAC Address
Meaning	Logical address of a device on a network	Physical hardware address of the NIC
Size	32 bits (IPv4) / 128 bits (IPv6)	48 bits
Assigned by	Network / DHCP server / ISP	Manufacturer
Changes?	Can change when network changes	Fixed (can be spoofed by software)
Used for	Routing across different networks	Delivery within the local network
Layer	Network layer (Layer 3)	Data Link layer (Layer 2)

3.5 Router, Switch and Modem

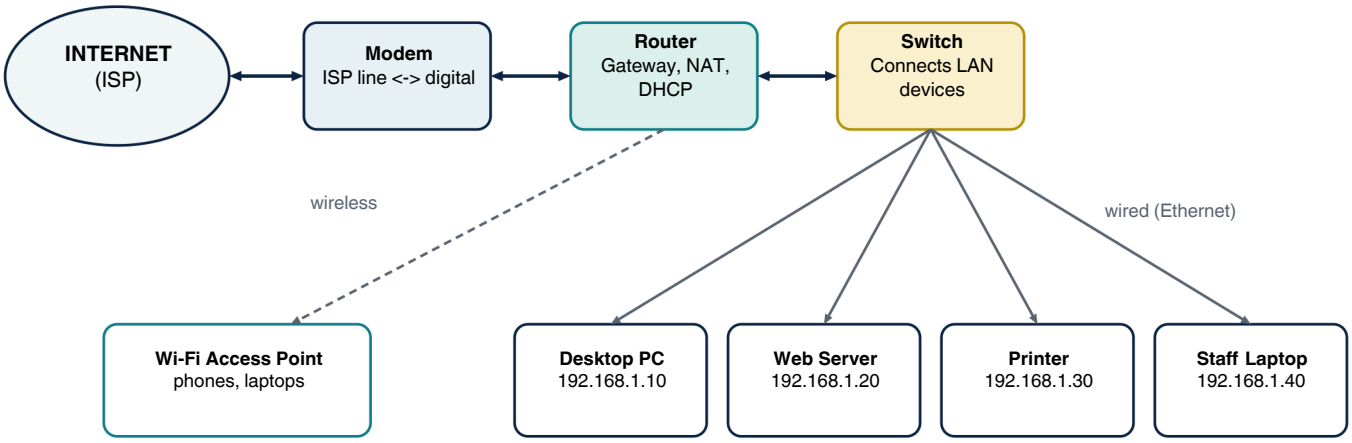


Figure 3: A simple college / office network showing modem, router, switch and connected devices

Modem	**Modulator-Demodulator.** Converts the digital signals of our network into signals suitable for the telephone, cable or fibre line of the ISP, and converts incoming signals back to digital. It is the device that connects the network to the Internet service provider.
Router	Connects **different networks** together (for example the LAN and the Internet) and forwards packets to their destination using **IP addresses** and routing tables. Home / campus routers also provide DHCP (automatic IP assignment), NAT, Wi-Fi and basic firewall features. Works at Layer 3.
Switch	Connects multiple devices **within the same LAN**. It learns the MAC address of each device on each port and forwards data only to the port of the intended device, unlike a hub which broadcasts to everyone. It makes the LAN faster and more secure. Works at Layer 2.

Device	OSI layer	Uses	Connects
Modem	Physical (Layer 1)	Signal conversion	Network to ISP
Switch	Data Link (Layer 2)	MAC address	Devices inside a LAN
Router	Network (Layer 3)	IP address	One network to another network

3.6 DNS (Domain Name System)

The**Domain Name System (DNS)** is the phonebook of the Internet. Humans remember names like www.college.edu.in, but computers communicate using IP addresses. DNS automatically translates the domain name into the IP address of the server. DNS is a distributed, hierarchical system made of **root servers, TLD servers (.com, .in, .edu) and authoritative name servers**. To make browsing faster, answers are cached for a time called the TTL (Time To Live).

DNS record	Purpose	Example
A	Maps a domain to an IPv4 address	college.edu.in -> 203.0.113.25
AAAA	Maps a domain to an IPv6 address	college.edu.in -> 2001:db8::25
CNAME	Alias of another domain name	www -> college.edu.in
MX	Mail server of the domain	Mail for @college.edu.in

Task 4: Internet Communication Process (15 Marks)

This section explainswhathappens,stepbystep,whenastudenttypes
the browser and presses Enter.

4.1 Client and Server

- **Client:** the device or software that asks for a service, here the student's computer and its web browser (Chrome, Firefox, etc.).
- **Server:** a powerful computer that stores the website and provides the service, here the college's web server (software such as Apache / Nginx plus a database).
- This model is called the **client-server model**: many clients send requests, and the server sends back responses.

4.2 Request and Response

Communication on the web follows a **request-response cycle**. The browser sends an **HTTP request** (for example GET /admissions); the server processes it and sends back an **HTTP response** containing a status code and the content (HTML, CSS, images).

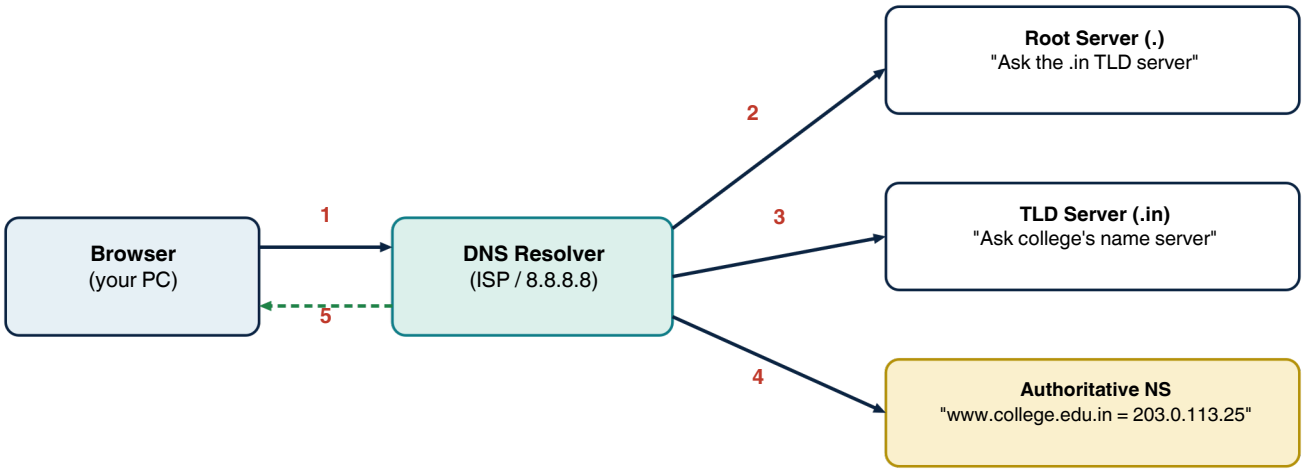
Item	Examples
Common request methods	GET (fetch a page), POST (submit a form, e.g. login), PUT / DELETE (update / remove data)
Common status codes	200 OK, 301 Moved Permanently, 403 Forbidden, 404 Not Found, 500 Internal Server Error

4.3 HTTP and HTTPS

Feature	HTTP	HTTPS
Full form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Default port	80	443
Encryption	None, data travels as readable text	Encrypted using TLS (SSL)
Certificate	Not required	Requires a digital certificate issued by a Certificate Authority
Protection	Data can be read or changed by attackers on the path	Confidentiality, integrity and server authentication
Browser indicator	"Not secure" warning	Padlock icon

4.4 DNS Lookup

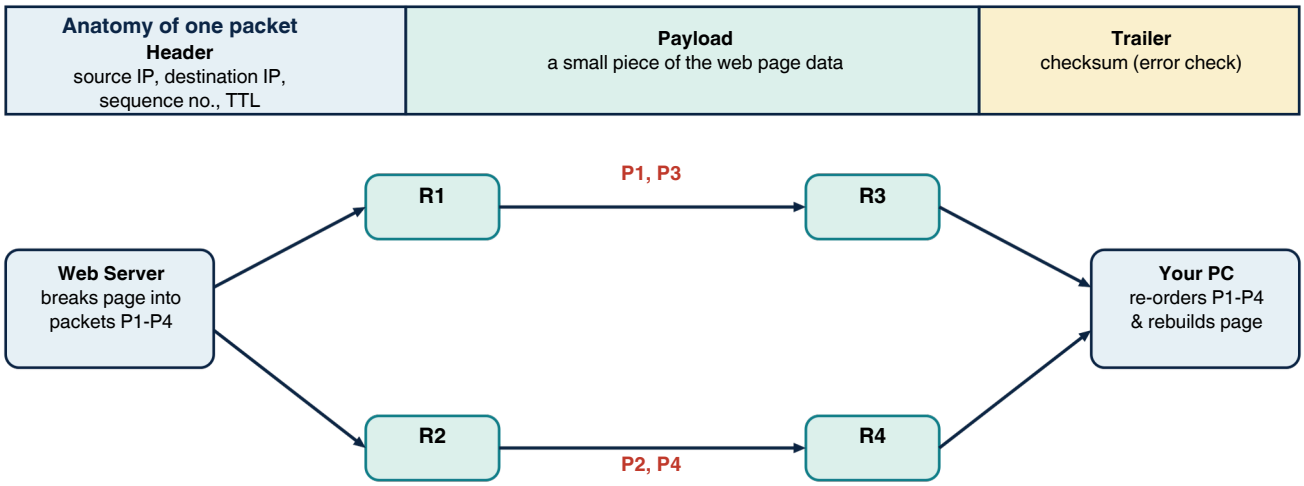
Before the browser can contact the server, it must find the server's IP address. The lookup order is: **(1)** browser cache, **(2)** operating-system cache / hosts file, **(3)** the DNS resolver of the ISP or a public DNS (e.g. 8.8.8.8), which, if it does not already know the answer, asks the **(4)** root server, **(5)** the .in TLD server and **(6)** the college's authoritative name server. The IP address is returned and cached.



Result: browser now knows the web server is at 203.0.113.25 (example IP).

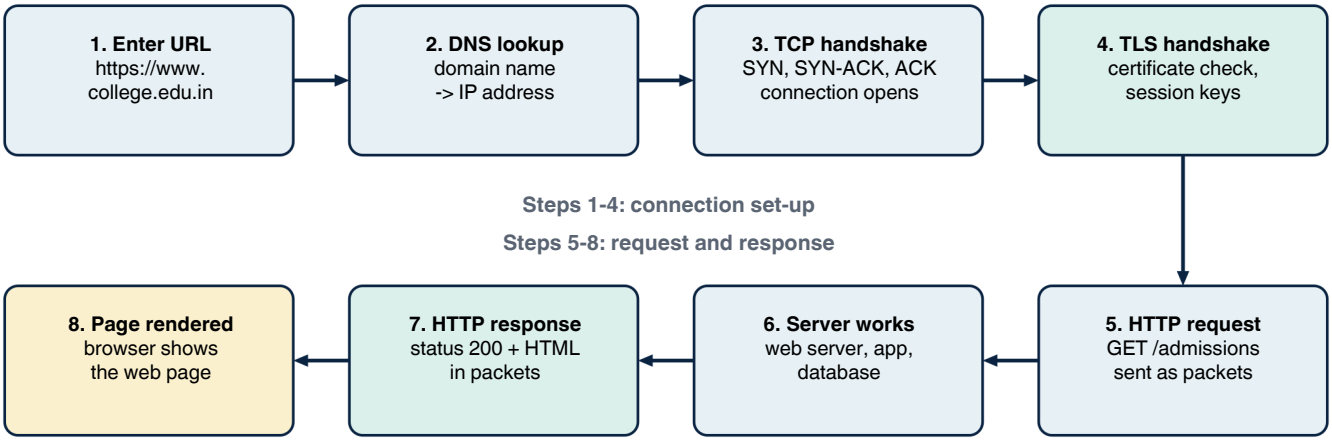
4.5 Packet Transmission

Data is never sent in one big piece. The TCP/IP protocols break it into small units called **packets** (usually up to about 1500 bytes each). Each packet carries a header (source and destination IP address, sequence number, TTL), a payload (part of the data) and error-checking information. Routers forward each packet hop by hop towards the destination; different packets may take different paths. At the destination, TCP checks for missing packets (requesting a resend if needed), puts them in the right order and rebuilds the original data.



TCP/IP layer	Protocols	Role in loading the website
Application	HTTP / HTTPS, DNS	Browser and server exchange web pages; DNS finds the IP
Transport	TCP (port 443)	Reliable delivery, splitting into segments, ordering, retransmission
Internet	IP	Adds source / destination IP addresses; routers choose the path
Network access	Ethernet / Wi-Fi (MAC addresses)	Sends frames over the physical link to the next device

4.6 Flow Diagram: From Your Computer to the Website



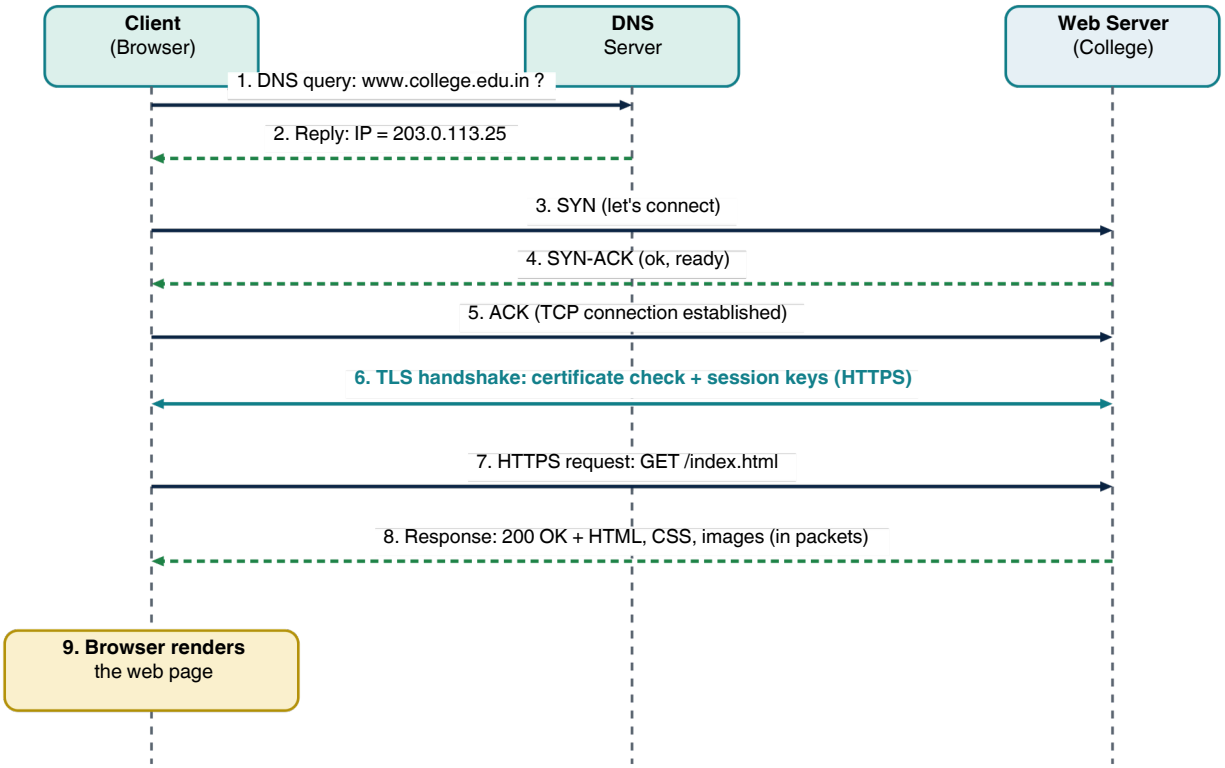


Figure 7: Message sequence between client, DNS server and web server

Summary in words

Thestudent types the URL; the browser finds the IP address using DNS; a TCP connection is opened with a three-way handshake; for HTTPS a TLS handshake authenticates the server and creates encryption keys; the browser sends an HTTP request split into packets; the server processes it (with the application and database) and returns an HTTP response as packets; finally the browser reassembles the data and displays the page. All of this happens in a fraction of a second.

Task 5: Network Security Basics (15 Marks)

Network security is the set of toolsand practicesthat protecttheconfidentiality, integrity and availability of data travelling across a network. No single tool is enough, so organisations use **defence in depth** (many layers of protection). Five basic building blocks are explained below.

5.1 Firewall

What it is	A hardware device or software that monitors incoming and outgoing network traffic and **allows or blocks it based on security rules** (IP addresses, ports, protocols).
How it works	It sits between a trusted network (college LAN) and an untrusted network (Internet) and inspects each packet. Types: packet-filtering, stateful inspection, application / proxy firewall and Next-Generation Firewall (NGFW).
Why important	Blocks unauthorised access, port scans and many attacks, controls which services are reachable (e.g. only ports 80/443 for the website) and logs suspicious activity.
College example	The firewall allows students to browse HTTPS sites but blocks traffic from known malicious IPs to the ERP server.

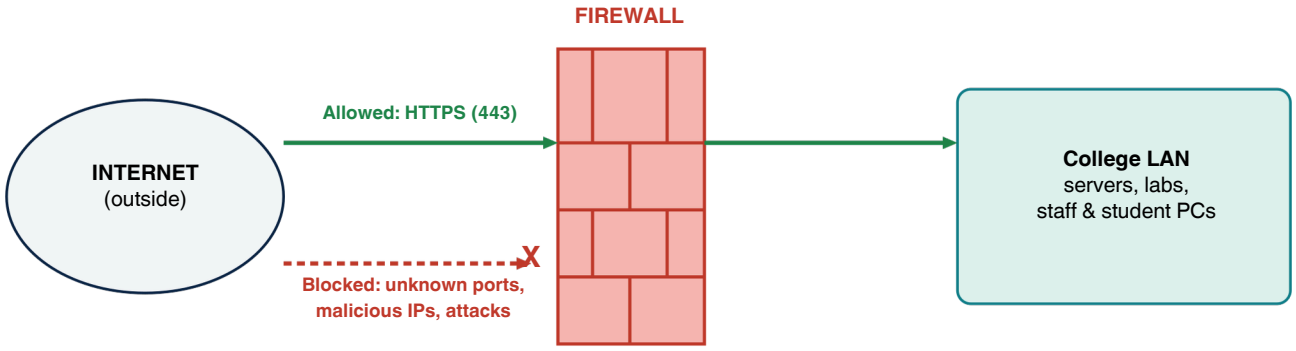


Figure 8: A firewall filtering traffic between the Internet and the college LAN

5.2 Antivirus

What it is	Software that **detects, blocks and removes** malicious programs such as viruses, worms, trojans and spyware.
How it works	Uses signature-based detection (known malware patterns), heuristic / behaviour analysis (suspicious actions) and cloud-based intelligence; performs real-time and scheduled scans and quarantines infected files.
Why important	Protects devices from malware infections, prevents data loss and stops infected machines from spreading malware to the rest of the network. It must be updated frequently to recognise new threats.
College example	Antivirus on lab computers automatically scans every pen drive inserted by students.

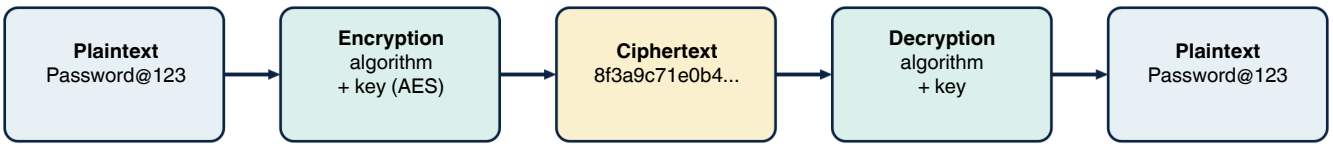
5.3 VPN (Virtual Private Network)

What it is	A service that creates a **secure, encrypted tunnel** over the Internet between a user's device and a VPN server, hiding the traffic and the real IP address.
How it works	The device encrypts data before sending it through the tunnel; the VPN server decrypts it and forwards it to the destination. Websites see the VPN server's IP instead of the user's.
Why important	Protects data on public Wi-Fi, allows safe remote access to private resources and improves privacy against eavesdropping. (A VPN does not make a user immune to malware or phishing.)
College example	A faculty member working from home connects to the college's VPN to reach internal servers securely.



5.4 Encryption

What it is	The process of converting readable data (**plaintext**) into unreadable coded data (**ciphertext**) using an algorithm and a key, so only authorised parties can read it.
How it works	**Symmetric encryption** (e.g. AES) uses the same key to encrypt and decrypt and is fast. **Asymmetric encryption** (e.g. RSA, ECC) uses a public key to encrypt and a private key to decrypt. HTTPS uses asymmetric encryption to exchange keys and symmetric encryption for the actual data.
Why important	Keeps data confidential even if it is intercepted or stolen, protects passwords and payments, and supports integrity through hashing and digital signatures.
College example	Fee payment pages use TLS encryption so card details cannot be read on the network; student passwords are stored as salted hashes.



Only someone holding the correct key can turn ciphertext back into readable plaintext.

5.5 Authentication

What it is	The process of **verifying the identity** of a user, device or system before granting access. (Authorisation then decides what the verified user is allowed to do.)
How it works	Users prove their identity with one or more factors: **something you know** (password, PIN), **something you have** (OTP, smart card, phone) and **something you are** (fingerprint, face). Using two or more factors is called **multi-factor authentication (MFA)**.
Why important	Stops unauthorised people from entering accounts and systems. Even if a password is stolen by phishing, MFA can still block the attacker.
College example	Students log in to the ERP with a username, password and an OTP sent to their registered mobile number.

5.6 Summary Table

Tool	Protects against	Main benefit
Firewall	Unauthorised network access, port scans	Controls what traffic can enter / leave
Antivirus	Viruses, malware, spyware	Detects and removes malicious software
VPN	Eavesdropping on public / untrusted networks	Private, encrypted connection
Encryption	Data theft and interception	Keeps data unreadable to outsiders
Authentication	Unauthorised account access	Confirms that users are who they claim to be

Task 6: Practical Activity (Option B) (15 Marks)

Activity chosen: OptionB-findtheIPaddressofmysystemandexplain IPv4, Public IP, Private IP and Dynamic vs Static IP.

6.1 Procedure to Find the IP Address

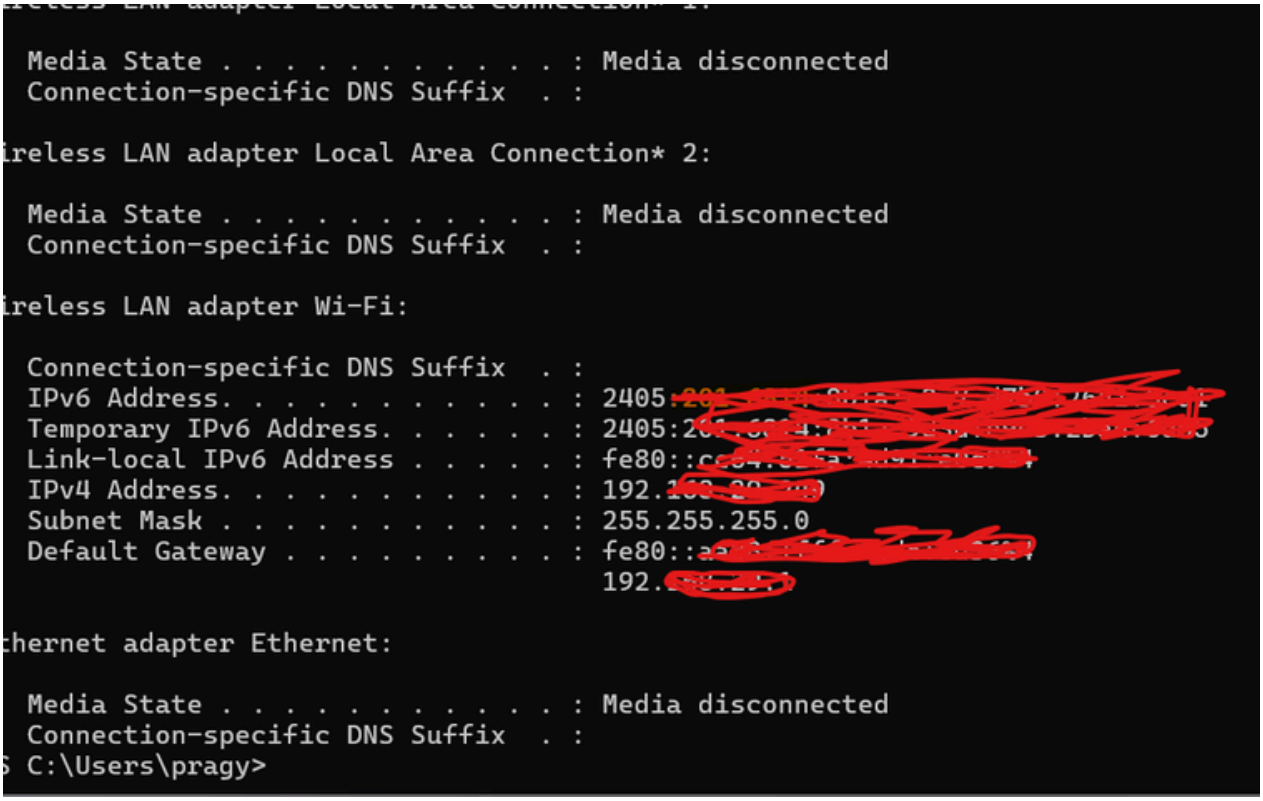
A. Private (local) IP address

Operating system	Steps
Windows	Press **Win + R**, type cmd and press Enter. Type ipconfig and press Enter. Note the **IPv4 Address**, **Subnet Mask** and **Default Gateway**. For more details (MAC address, DHCP status, lease time) use ipconfig /all.
Linux	Open the terminal and run ip a or hostname -I.
macOS	Run ipconfig getifaddr en0 in Terminal, or open System Settings > Network > Wi-Fi > Details.
Android / iPhone	Settings > Wi-Fi > tap the connected network (i / gear icon) > IP address.

B. Public IP address

Open a browser and search "what is my IP" or visit a site such as whatismyip.com, or run `curl ifconfig.me` in a terminal. The address displayed is the public IP assigned by the Internet Service Provider.

Screenshots of my system



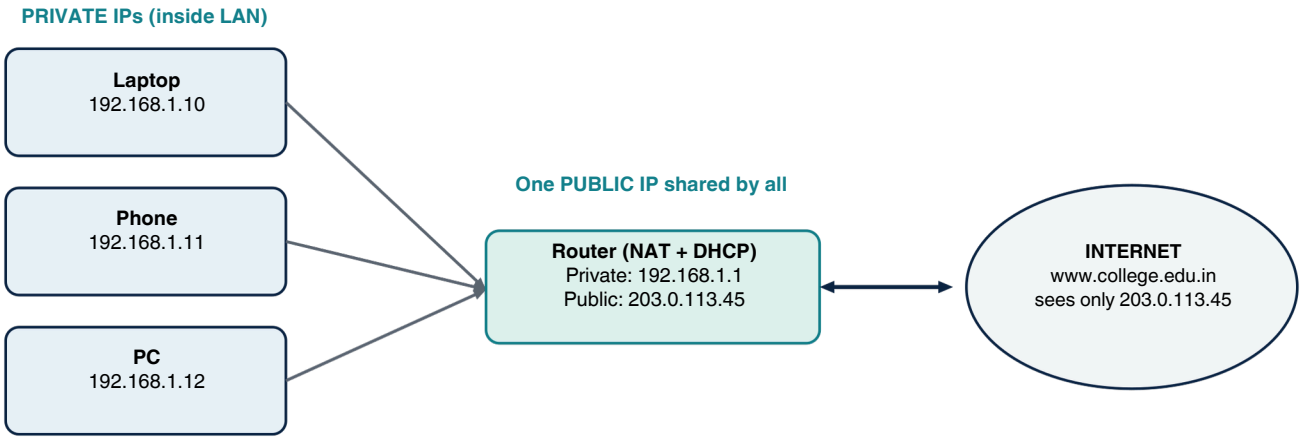
6.3 Public IP vs Private IP

Feature	Public IP	Private IP
Visibility	Visible and reachable on the Internet	Visible only inside the local network
Assigned by	Internet Service Provider (ISP)	Router (DHCP) or network administrator
Uniqueness	Globally unique	Can be reused in different private networks
Cost	Provided by the ISP, may cost extra for a fixed one	Free
Example	203.0.113.45 (example)	192.168.1.10
Security	More exposed to outside attacks	Hidden behind the router / NAT

Private IP ranges (RFC 1918):

Range	CIDR	Typical use
10.0.0.0 - 10.255.255.255	10.0.0.0/8	Large organisations, universities
172.16.0.0 - 172.31.255.255	172.16.0.0/12	Medium networks
192.168.0.0 - 192.168.255.255	192.168.0.0/16	Home and small office routers

How they work together (NAT): all devices inside the LAN use private IPs. When they access the Internet, the router performs **Network Address Translation (NAT)**, replacing the private address with its single public IP, so a website sees only the public address.



6.4 Dynamic IP vs Static IP

Feature	Dynamic IP	Static IP
Meaning	Automatically assigned for a limited time (lease) by a DHCP server and may change	Manually configured and remains fixed until changed
Management	Easy, no manual setup	Needs manual configuration and tracking
Cost	Cheaper (common for home users)	Usually costlier from the ISP
Reliability for hosting	Not ideal; address can change	Ideal for web servers, mail servers, CCTV, printers, remote access
Security	Harder to target continuously	Easier to target as the address never changes, needs a firewall
Example	Student laptop on campus Wi-Fi	College web server or network printer

Conclusion

Throughthis assignment I learnt that cyber security is about protecting the confidentiality, integrity and availability of information, and that threats such as malware, viruses, phishing, spyware and DoS attacks affect real platforms like a college website every day. I also learnt how networks are built from devices like modems, routers and switches, how addresses (IP and MAC) and DNS make communication possible, and how a request travels as packets from a browser to a web server and back. Finally, I understood how firewalls, antivirus, VPNs, encryption and authentication work together as layers of defence, and I practised finding and interpreting the IP address of my own system.

The most important lesson is that technology alone is not enough. Safe habits such as using strong unique passwords, enabling MFA, updating software and thinking before clicking links are equally important for keeping a digital system secure.